

Practical – 3

Practical Title: Responsive UI Design using CSS Grid, Flexbox, and Bootstrap/Tailwind

Problem Definition : Design responsive layouts for Home, About, Registration, Dashboard, and Events pages using CSS Grid and Flexbox. Optional to use CSS framework such as Bootstrap or Tailwind. Ensure mobile-first responsiveness.

A. Theory

1. How does the layout adapt for mobile, tablet, and desktop screens?
The StudentHub layout uses **CSS media queries** to adapt to different screen sizes.
 - On **desktop**, cards are displayed in multiple columns using CSS Grid.
 - On **tablet**, the layout adjusts the content width and spacing for better viewing.
 - On **mobile**, the cards change to a single-column layout and the navigation becomes vertical.
 - The logo, text, forms, and tables also adjust to provide a **responsive and user-friendly interface**.
2. Which layout technique is used and why?
The StudentHub portal uses **both CSS Flexbox and CSS Grid**.
 - **Flexbox** is mainly used for the header, logo, title, and navigation menu. It helps align the logo and title properly and makes the navigation flexible.
 - **CSS Grid** is used for displaying cards on the Dashboard, Courses, and Assignments pages.
 - Grid allows the cards to be arranged in multiple columns on larger screens. On mobile screens, the Grid changes to a **single-column layout** using media queries. This makes the pages responsive without creating separate mobile pages.
 - Media queries are used to change the layout according to the screen width. External style.css file is used so the same styling can be applied to all pages.

Thus, Flexbox, Grid, and media queries make the StudentHub interface **responsive, organized, and reusable**.

3. Are typography, spacing, colors, and contrast readable?

I have used the following features to make typography spacing, coolors, and contrast readable:

- The StudentHub portal uses simple and readable **Arial font** throughout the pages.
- The heading sizes are different from normal text, which makes the content easy to understand.
- Proper **padding and margins** are used around sections, forms, cards, and tables.
- A dark blue color is used for the header and footer to give the portal a consistent look.
- The main content has a light background, which makes the text easier to read.
- Links and buttons have clear colors so that users can identify them easily.
- The text and background have sufficient **contrast**, especially in the navigation and headings.
- On smaller screens, the font sizes and spacing are adjusted using media queries.

Overall, the design is simple and suitable for a student portal.

These choices make the website **comfortable to read and easy to use**.

4. Is the framework used consistently without unnecessary inline CSS?

In this practical, I mainly used **CSS3 with Flexbox, Grid, and media queries** instead of using a CSS framework like Bootstrap or Tailwind.

- All the styling is written in the external **style.css** file.
- The same CSS file is linked to all 10 StudentHub HTML pages.
This keeps the design consistent across the entire website.
- I avoided unnecessary inline CSS inside the HTML pages.
- Flexbox and Grid classes such as .logo-container and .card-container are reused wherever required.
- Media queries are also written in one place to handle different screen sizes.
This makes the code easier to understand and modify.
- It also avoids repeating the same styling in every HTML file.

Therefore, the CSS is **organized, reusable, and consistent across the StudentHub portal**.

B. Implementation/ Output/Result:

The following CSS features/properties were used in the **StudentHub responsive UI**:

• CSS Feature / Property	• How it was used
• <code>* selector</code>	• Used to reset default margin, padding, and box sizing for all elements.
• <code>body</code>	• Used to set the overall font, background color, text color, and line spacing.
• <code>header</code>	• Used to style the StudentHub header with background color, text alignment, and spacing.
• <code>display: flex</code>	• Used for the logo/title area and navigation to arrange elements flexibly.
• <code>display: grid</code>	• Used to arrange cards in rows and columns on Dashboard, Courses, and Assignments pages.
• <code>flex-direction</code>	• Used to change navigation and logo layout to vertical on mobile screens.
• <code>justify-content</code>	• Used to horizontally position and distribute Flexbox elements.
• <code>align-items</code>	• Used to vertically align the logo and title.
• <code>flex-wrap</code>	• Allows navigation links to move to another line when necessary.
• <code>grid-template-columns</code>	• Controls the number of columns in the card layout.
• <code>gap</code>	• Adds consistent spacing between navigation links and cards.
• <code>@media queries</code>	• Used to create different layouts for mobile, tablet, and desktop screens.
• <code>width / max-width</code>	• Used to control the content width on different screen sizes.
• <code>margin</code>	• Used to create space outside elements and center content.
• <code>padding</code>	• Used to create space inside sections, forms, buttons, and other elements.
• <code>background-color</code>	• Used to provide colors to the header, footer, cards, sections, and page background.
• <code>color</code>	• Used to set readable text and link colors.
• <code>font-family</code>	• Used to set Arial as the main font.

• CSS Feature / Property	• How it was used
• <code>font-size</code>	• Used to control heading, navigation, form, and mobile text sizes.
• <code>font-weight</code>	• Used to make labels and links more prominent.
• <code>line-height</code>	• Used to improve readability by controlling line spacing.
• <code>text-align</code>	• Used to center the header, logo, and footer content.
• <code>text-decoration</code>	• Used to remove underlines from navigation links and add them on hover.
• <code>border</code>	• Used for form fields and table borders.
• <code>border-radius</code>	• Used to give rounded corners to sections, cards, buttons, and inputs.
• <code>border-left</code>	• Used to highlight article cards and the aside section.
• <code>box-shadow</code>	• Used to give sections a simple card-like appearance.
• <code>object-fit</code>	• Used to keep the StudentHub logo properly fitted within its dimensions.
• <code>:hover</code>	• Used to change the appearance of navigation links and buttons when the mouse is over them.
• <code>nth-child(even)</code>	• Used to give alternate table rows a different background.
• <code>cursor: pointer</code>	• Used to show that buttons are clickable.
• <code>resize: vertical</code>	• Allows the contact message textarea to be resized vertically.
• <code>border-collapse</code>	• Used to make table borders appear as a single border.

C. Evidence of Work (Screenshots):

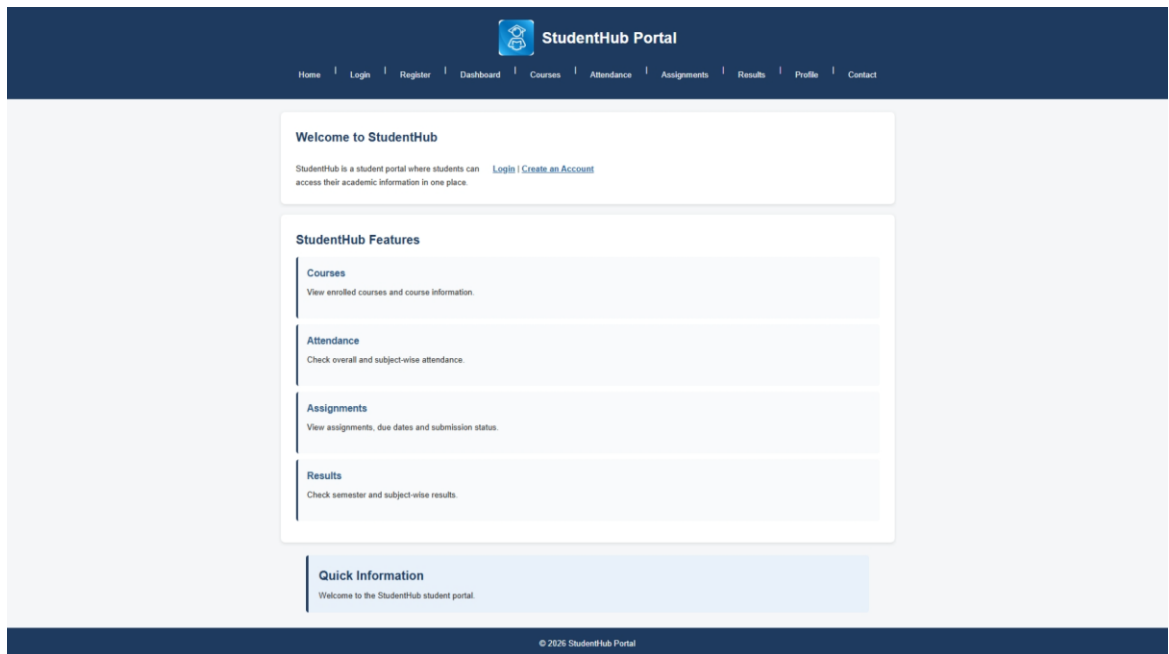


Fig1: Index

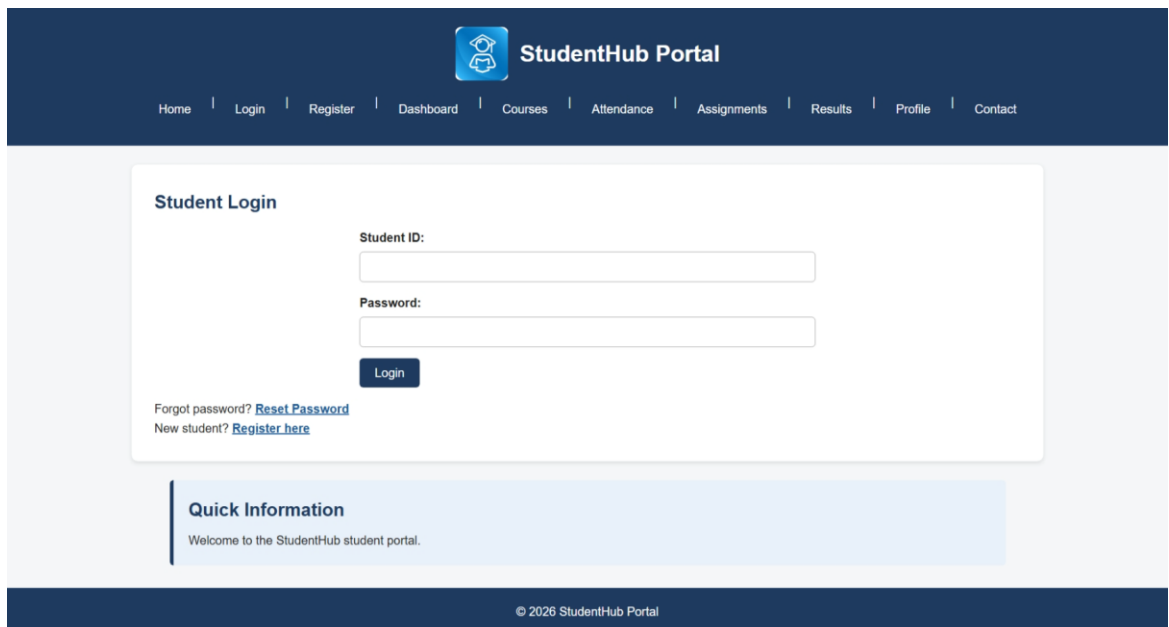
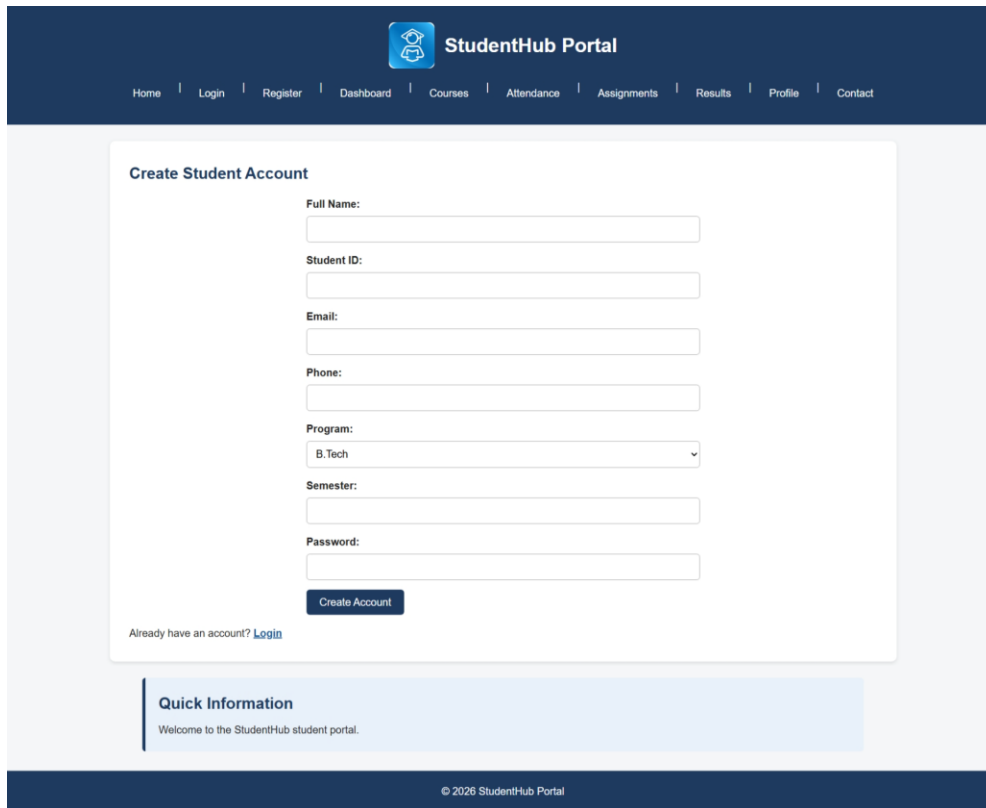


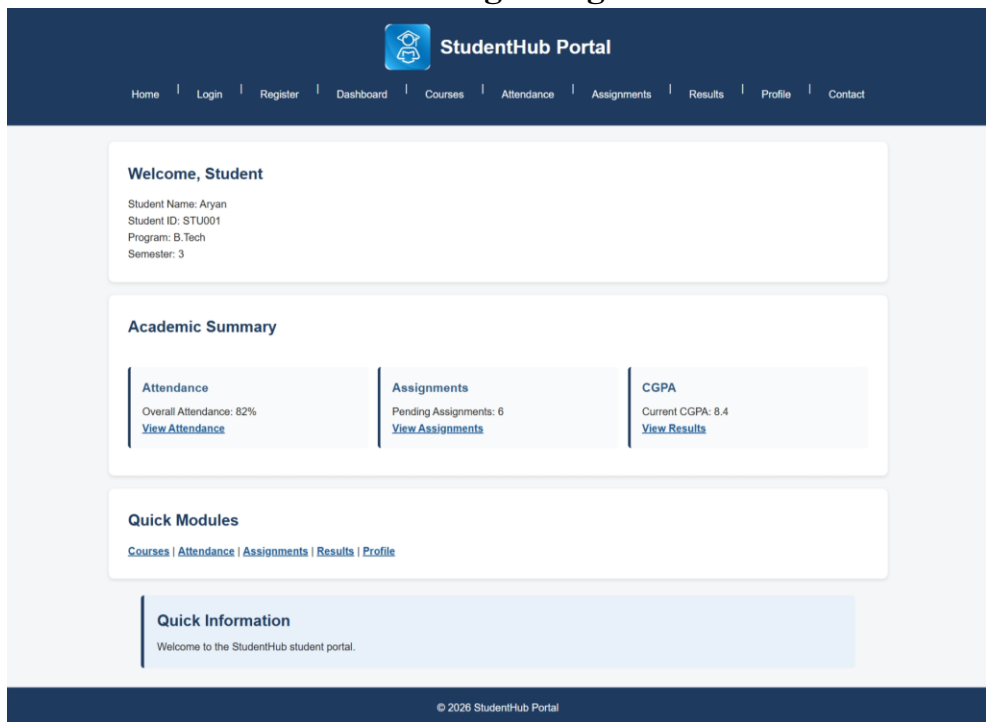
Fig2: Login



The image shows the 'Create Student Account' form on the StudentHub Portal. The form is located within a white box on a light gray background. The header of the page is dark blue with the 'StudentHub Portal' logo and a navigation menu. The form fields are as follows:

- Full Name:** Text input field.
- Student ID:** Text input field.
- Email:** Text input field.
- Phone:** Text input field.
- Program:** Dropdown menu with 'B.Tech' selected.
- Semester:** Text input field.
- Password:** Text input field.

Below the form is a 'Create Account' button. At the bottom of the form box, there is a link: 'Already have an account? [Login](#)'. Below the form box is a 'Quick Information' section with the text 'Welcome to the StudentHub student portal.' The footer of the page is dark blue with the copyright notice '© 2026 StudentHub Portal'.

Fig3: Register

The image shows the 'Welcome, Student' dashboard on the StudentHub Portal. The dashboard is located within a white box on a light gray background. The header of the page is dark blue with the 'StudentHub Portal' logo and a navigation menu. The dashboard content is as follows:

- Welcome, Student**
 - Student Name: Aryan
 - Student ID: STU001
 - Program: B.Tech
 - Semester: 3
- Academic Summary**

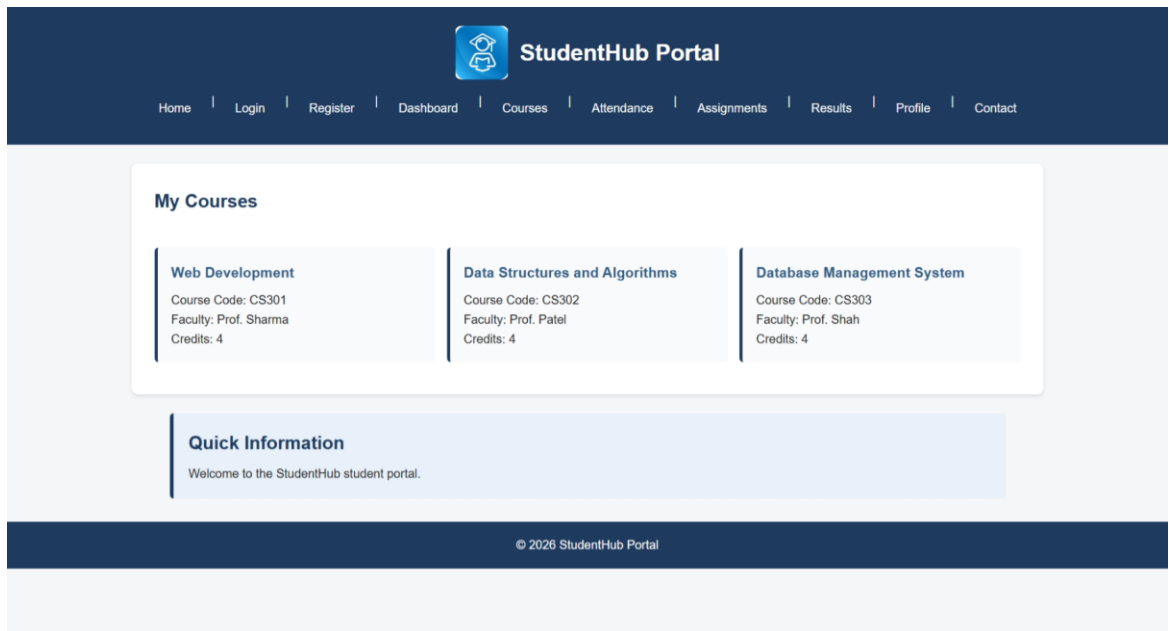
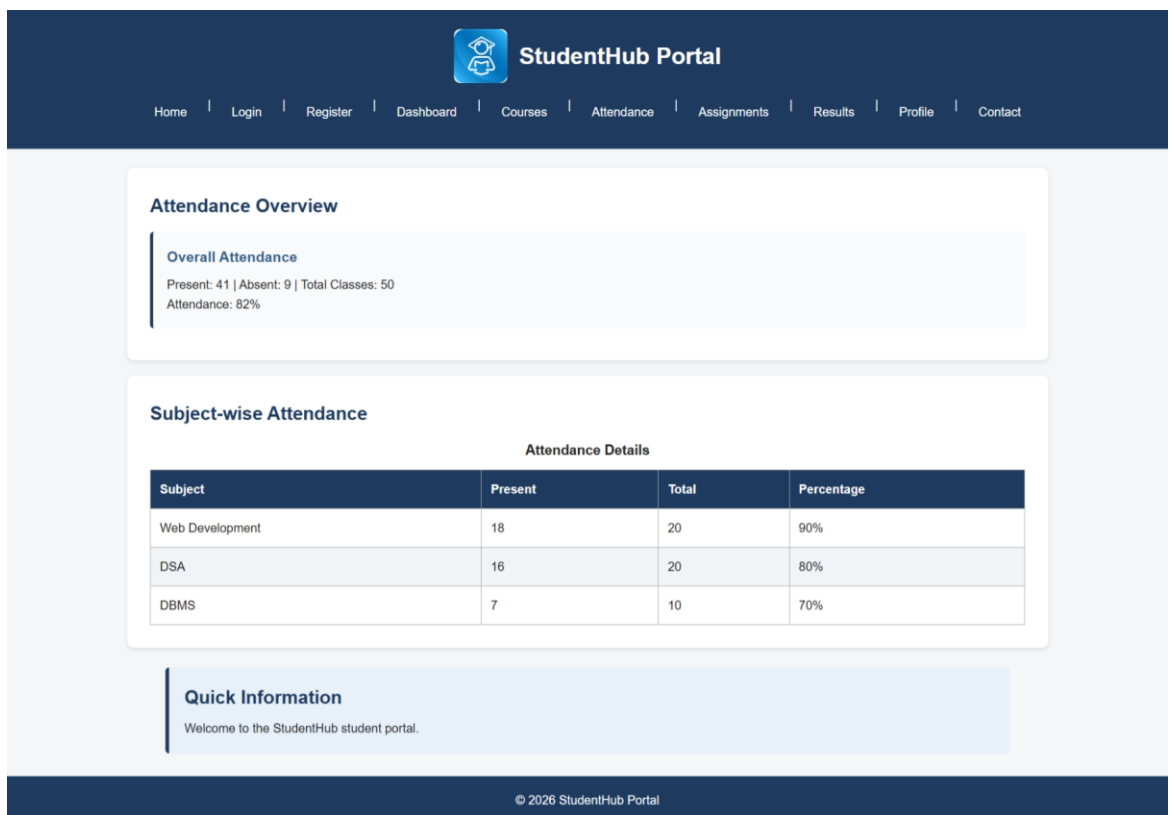
Attendance	Assignments	CGPA
Overall Attendance: 82% View Attendance	Pending Assignments: 6 View Assignments	Current CGPA: 8.4 View Results
- Quick Modules**

[Courses](#) | [Attendance](#) | [Assignments](#) | [Results](#) | [Profile](#)
- Quick Information**

Welcome to the StudentHub student portal.

The footer of the page is dark blue with the copyright notice '© 2026 StudentHub Portal'.

Fig4: Dashboard

**Fig5: Courses****Fig6: Attendance**

The screenshot shows the StudentHub Portal interface. At the top is a dark blue header with the StudentHub logo and the text 'StudentHub Portal'. Below the header is a navigation bar with links: Home, Login, Register, Dashboard, Courses, Attendance, Assignments, Results, Profile, and Contact. The main content area has a light gray background. It features a section titled 'Assignment List' with three cards. The first card is 'Practical 1 - Requirement Analysis' for 'Web Development' with a due date of 10 August 2026 and status 'Pending'. The second card is 'Practical 2 - Semantic HTML' for 'Web Development' with a due date of 20 August 2026 and status 'Pending'. The third card is 'DSA Lab 2' for 'DSA' with a due date of 12 August 2026 and status 'Submitted'. Below this is a section titled 'Assignment Actions' with links 'View Assignment' and 'Submit Assignment'. At the bottom of the main content area is a 'Quick Information' box with the text 'Welcome to the StudentHub student portal.' The footer is a dark blue bar with the text '© 2026 StudentHub Portal'.

StudentHub Portal

Home | Login | Register | Dashboard | Courses | Attendance | Assignments | Results | Profile | Contact

Assignment List

Assignment Title	Course	Due Date	Status
Practical 1 - Requirement Analysis	Web Development	10 August 2026	Pending
Practical 2 - Semantic HTML	Web Development	20 August 2026	Pending
DSA Lab 2	DSA	12 August 2026	Submitted

Assignment Actions

[View Assignment](#) | [Submit Assignment](#)

Quick Information

Welcome to the StudentHub student portal.

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Fig7: Assignment

The screenshot shows the StudentHub Portal interface for the Semester Result page. It has the same header and navigation bar as Fig7. The main content area has a light gray background. It features a section titled 'Semester Result' with a card for 'Current Semester' showing 'Semester: 3', 'SGPA: 8.6', and 'CGPA: 8.4'. Below this is a section titled 'Subject Results' with a table for 'Semester 3 Results'. The table has four columns: Subject, Internal, External, and Grade. It lists three subjects: Web Development (Internal: 42, External: 46, Grade: A), DSA (Internal: 40, External: 44, Grade: A), and DBMS (Internal: 41, External: 43, Grade: A). At the bottom of the main content area is a 'Quick Information' box with the text 'Welcome to the StudentHub student portal.' The footer is a dark blue bar with the text '© 2026 StudentHub Portal'.

StudentHub Portal

Home | Login | Register | Dashboard | Courses | Attendance | Assignments | Results | Profile | Contact

Semester Result

Current Semester

Semester: 3
SGPA: 8.6
CGPA: 8.4

Subject Results

Semester 3 Results

Subject	Internal	External	Grade
Web Development	42	46	A
DSA	40	44	A
DBMS	41	43	A

Quick Information

Welcome to the StudentHub student portal.

© 2026 StudentHub Portal

Fig8: Result

 **StudentHub Portal**

Home | Login | Register | Dashboard | Courses | Attendance | Assignments | Results | Profile | Contact

Student Profile

Personal Details
Name: Aryan Chauhan
Student ID: STU001
Email: student@example.com
Phone: 9876543210

Academic Details
Program: B.Tech
Department: Computer Engineering
Semester: 3

Profile Actions
[Edit Profile](#) | [Change Password](#)

Quick Information
Welcome to the StudentHub student portal.

© 2026 StudentHub Portal

Fig9: Profile

 **StudentHub Portal**

Home | Login | Register | Dashboard | Courses | Attendance | Assignments | Results | Profile | Contact

Contact StudentHub

If you have questions or problems with the portal, contact us using the form below.

Name:

Email:

Subject:

Message:

Quick Information
Welcome to the StudentHub student portal.

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Fig10: Contact

D. GitHub Repository Link:

[StudentHubPortal/css/style.css at main · aryan150806-wq/StudentHubPortal](https://StudentHubPortal/css/style.css)

E. Challenges Faced:

- Understanding responsive design – It was initially difficult to understand how the same webpage should adjust for mobile, tablet, and desktop screens.
- Using CSS Grid – Arranging cards into multiple columns and changing them to one column on mobile required practice.
- Using Flexbox – Aligning the logo, title, and navigation properly was a little challenging at first.
- Writing media queries – Deciding the appropriate screen sizes and CSS changes for different devices took some understanding.
- Maintaining consistent styling – Applying the same colors, fonts, spacing, and layout across all 10 pages required careful editing.
- Adding the logo – Setting the correct relative path for logo.jpg and styling it properly was initially confusing.
- Managing external CSS – Linking style.css correctly from the HTML folder and checking whether the CSS was loading was an important challenge.
- Testing different screen sizes – Checking the pages in mobile, tablet, and desktop views helped identify layout and spacing problems.

F. Learning Outcomes:

Through this practical, I learned how to make the StudentHub portal **responsive and user-friendly** using CSS3. I gained practical experience with **Flexbox, CSS Grid, media queries, spacing, typography, colors, borders, and responsive layouts**. I also learned how to create a consistent header, navigation, logo, cards, forms, tables, and footer across multiple pages.

I also understood the importance of keeping CSS in an **external style.css file** instead of using unnecessary inline styling. By testing the pages at different screen sizes, I learned how layouts behave on mobile, tablet, and desktop devices. Overall, this practical improved my understanding of **responsive web design, reusable CSS**,

layout techniques, and maintaining a consistent user interface across the StudentHub portal.

Faculty Remarks: